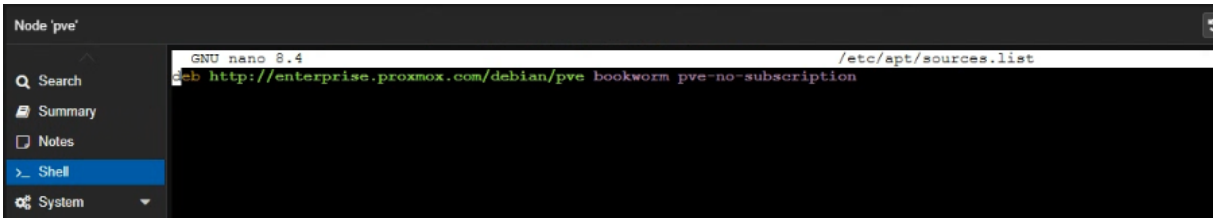
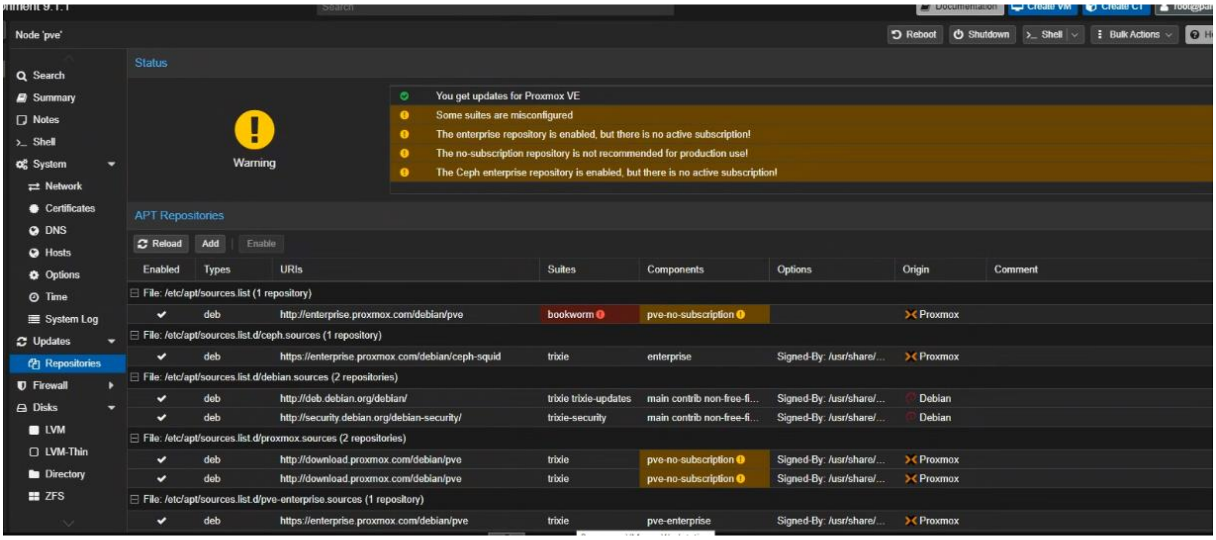
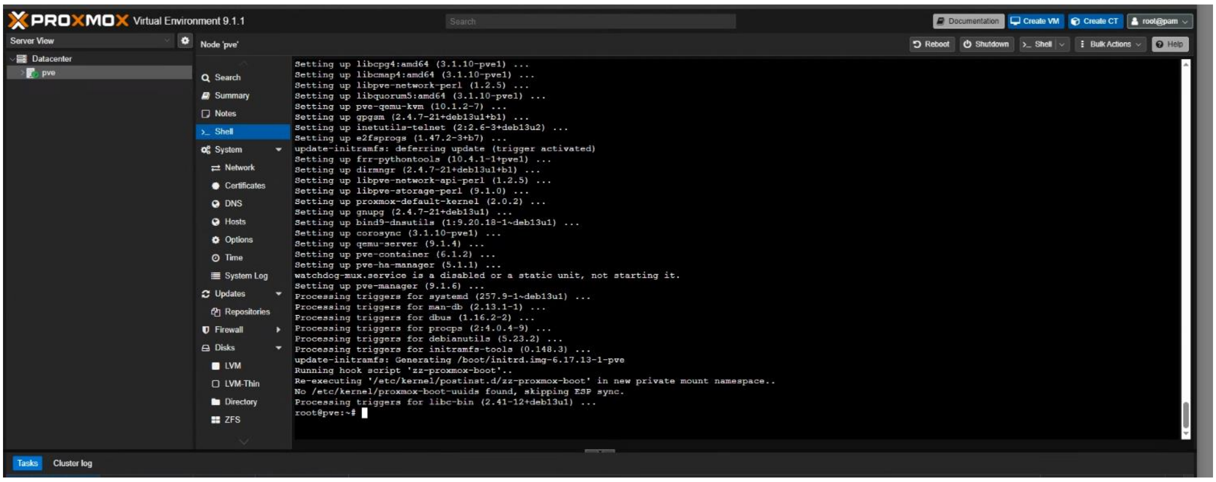
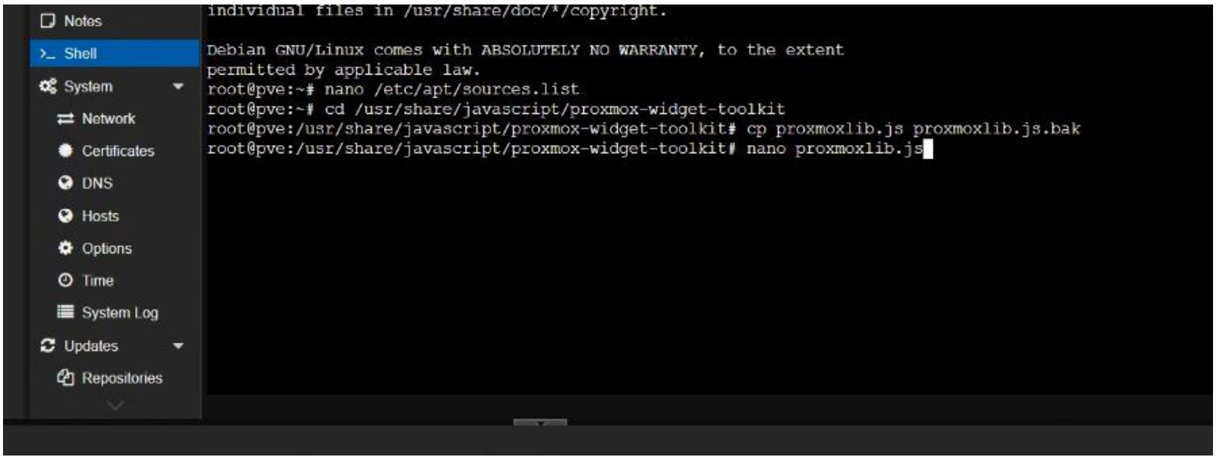
Zad 1.

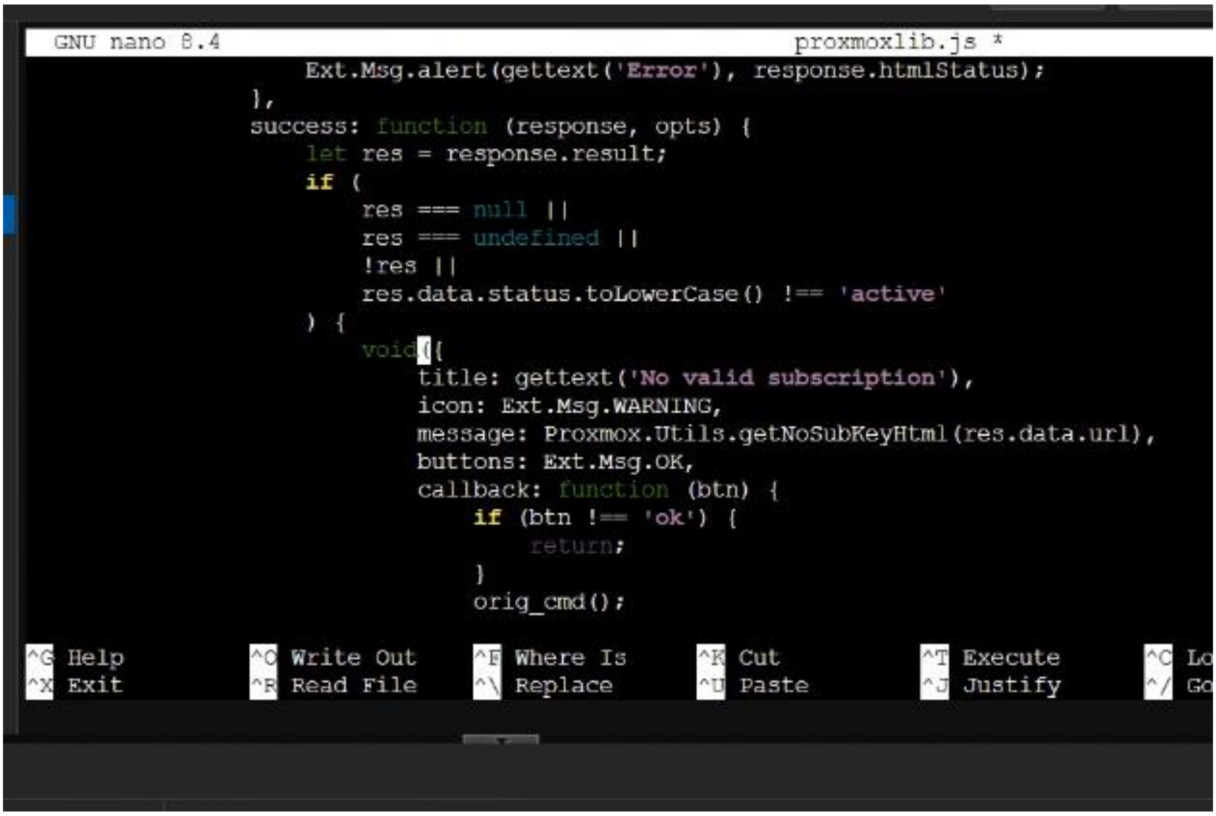






Wykonanie polecenia apt upgrade

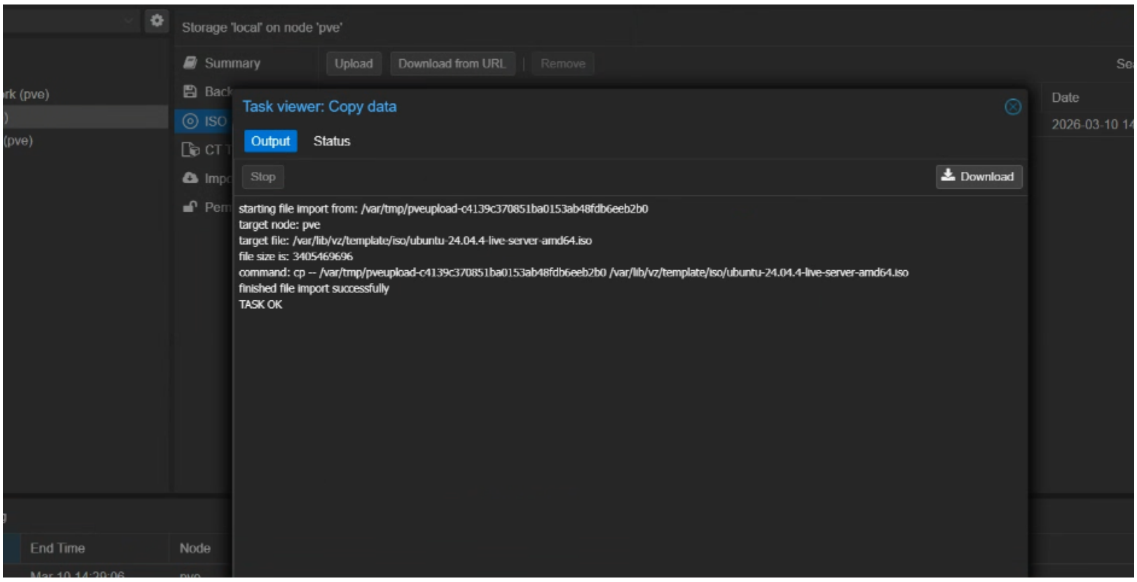


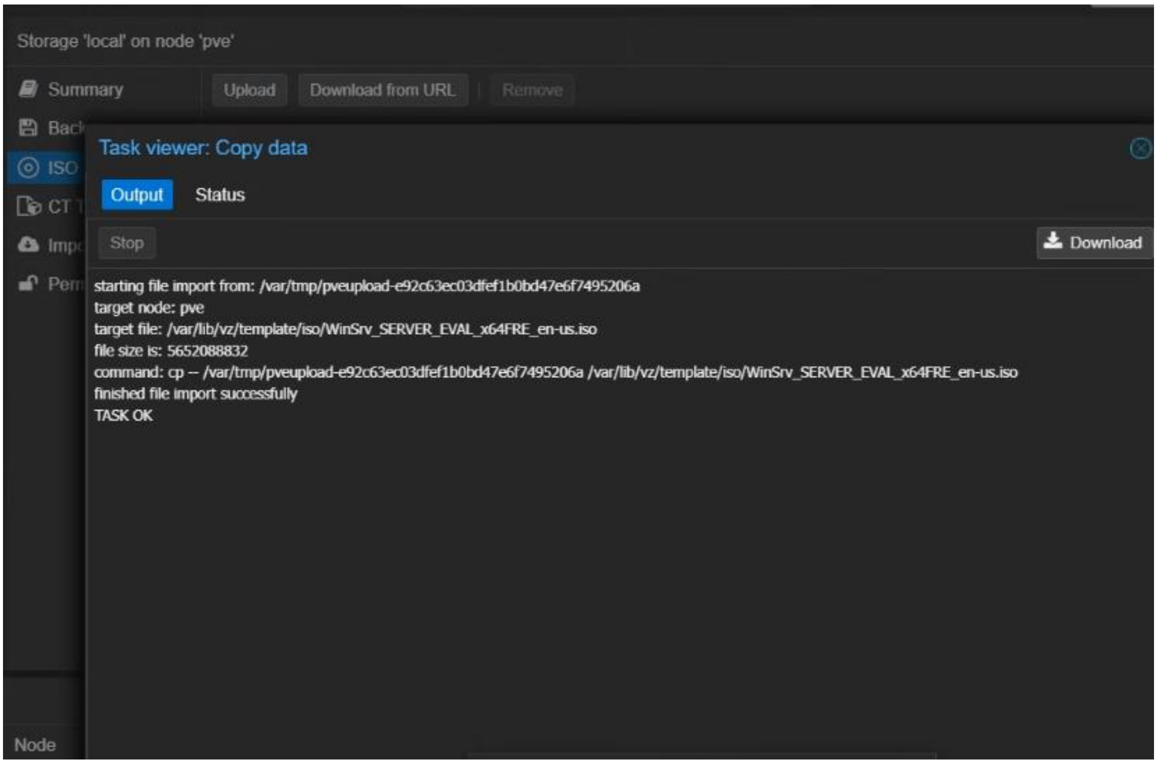


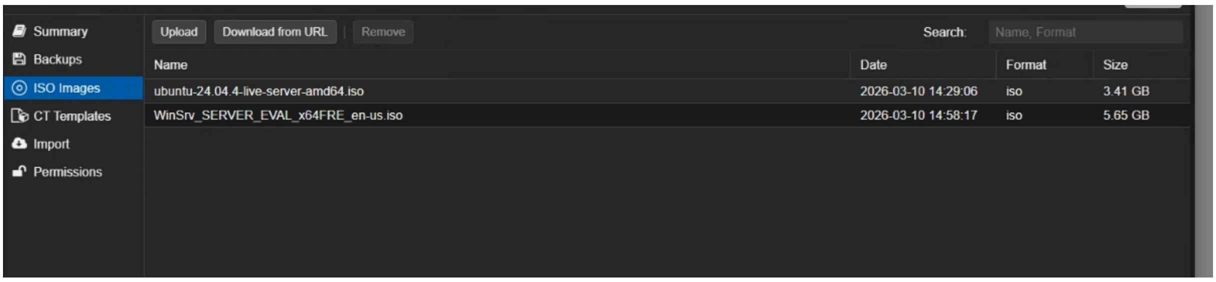
Po zalogowaniu nie występuje okienko. 😊

Zad 2.

Import obrazu ISO systemu Ubuntu Server 24.04 oraz Windows Server do lokalnego magazynu Proxmox.

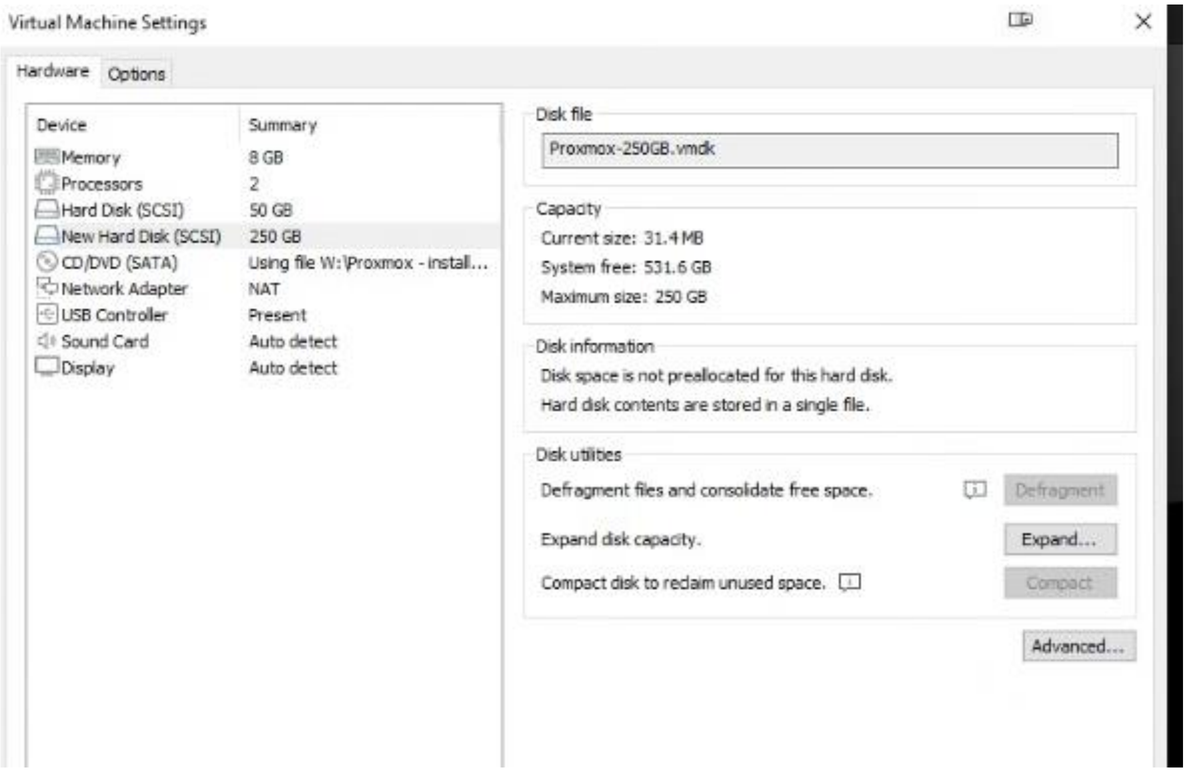






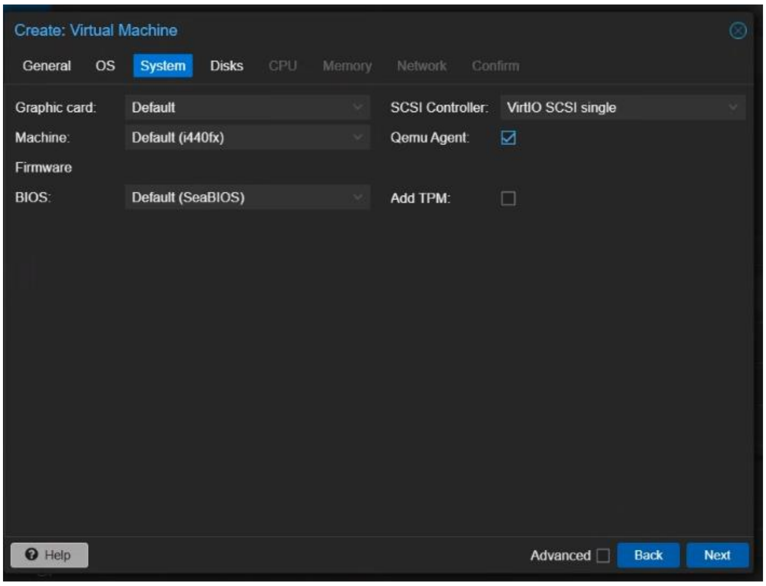
Zad 3.

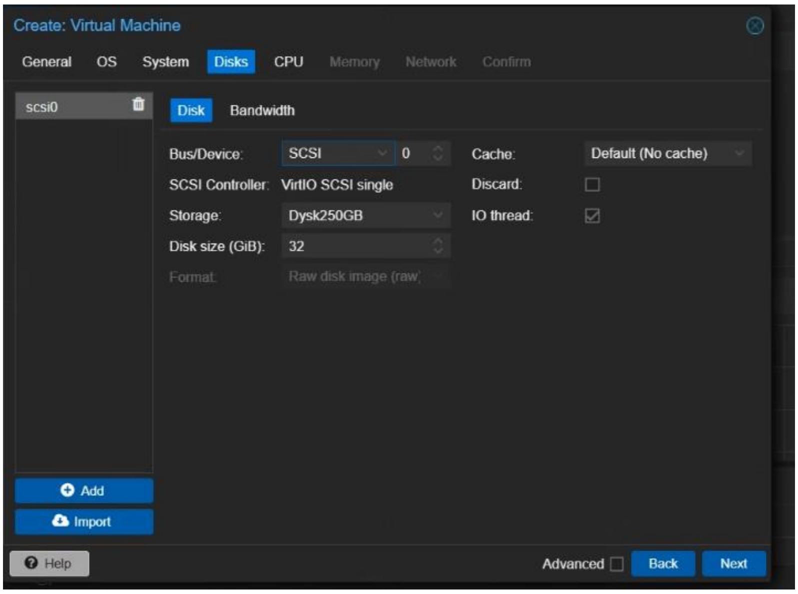
Dodanie nowego dysku o wielkości 250GB.

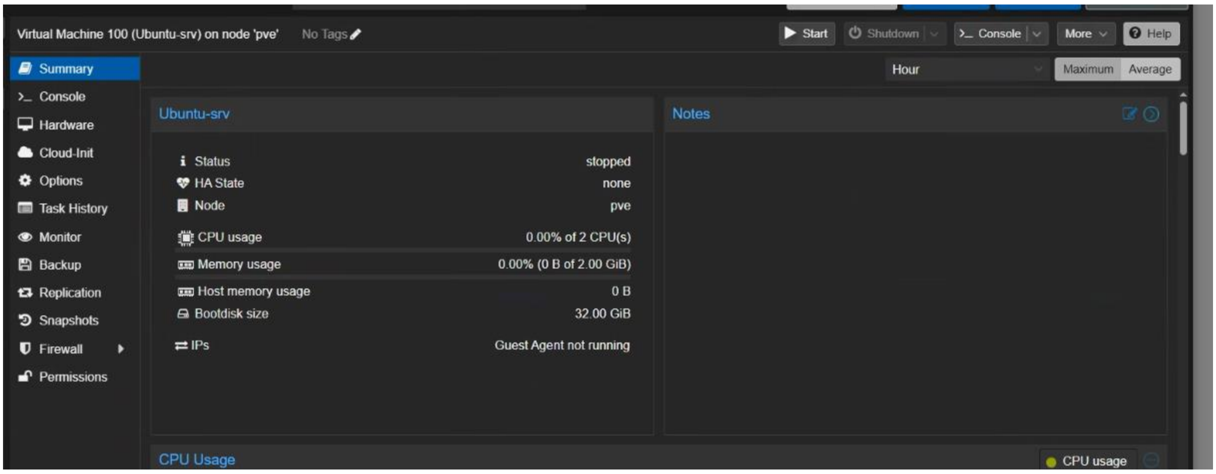


Zad 4.

Instalacja systemu Ubuntu Server.



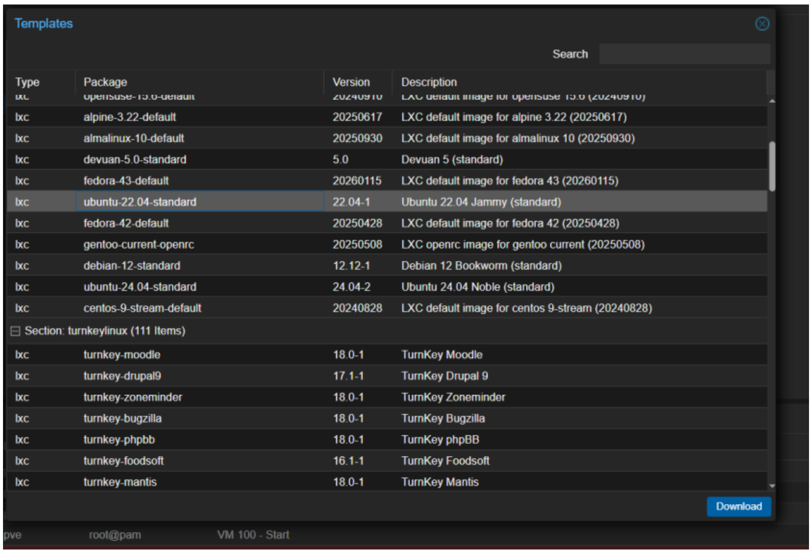


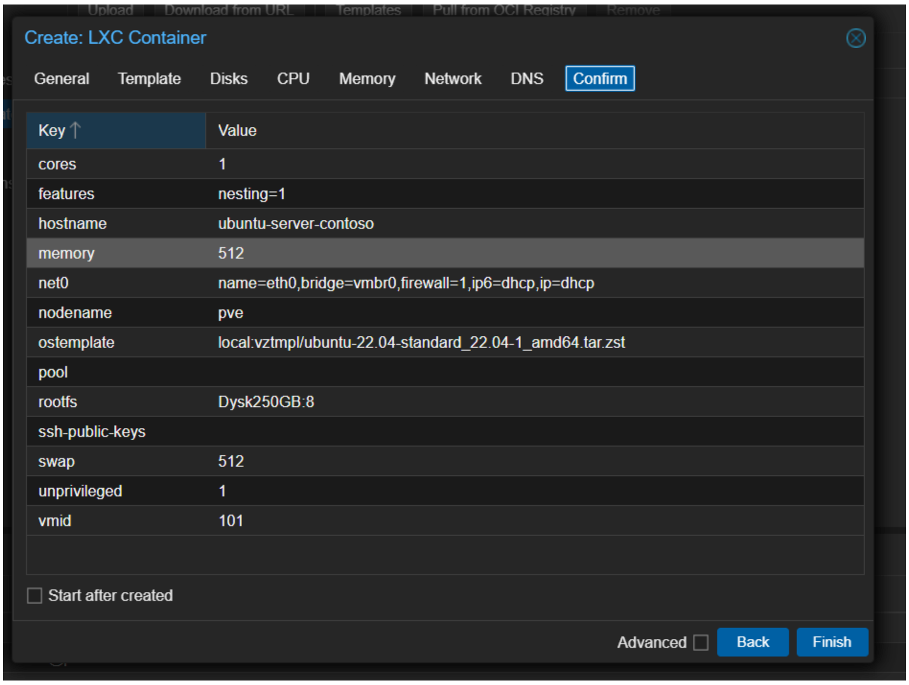


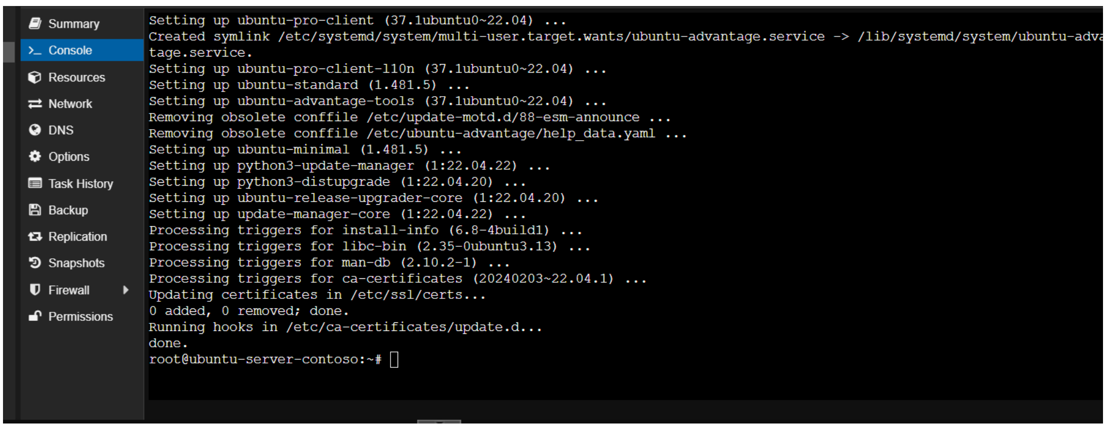
Kacper Śliwka

Zad 5.

Tworzenie kontenera z systemem Ubuntu Server.

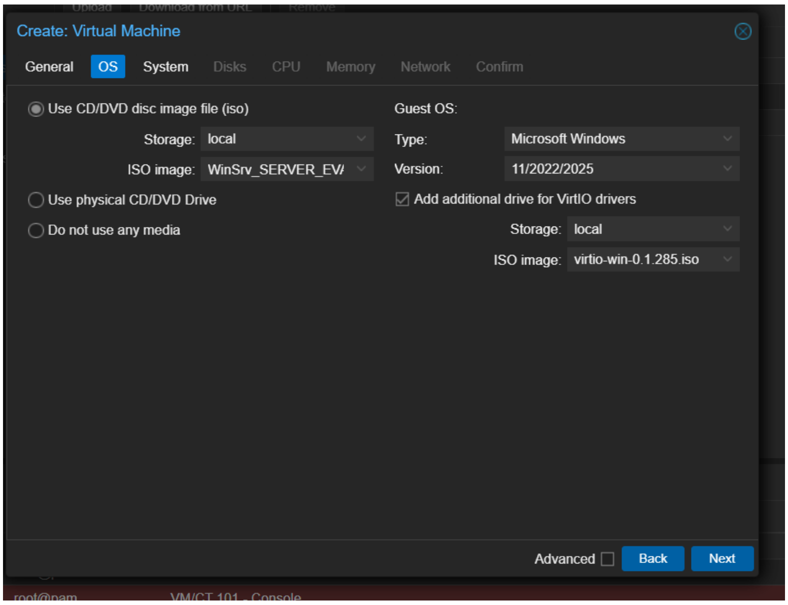


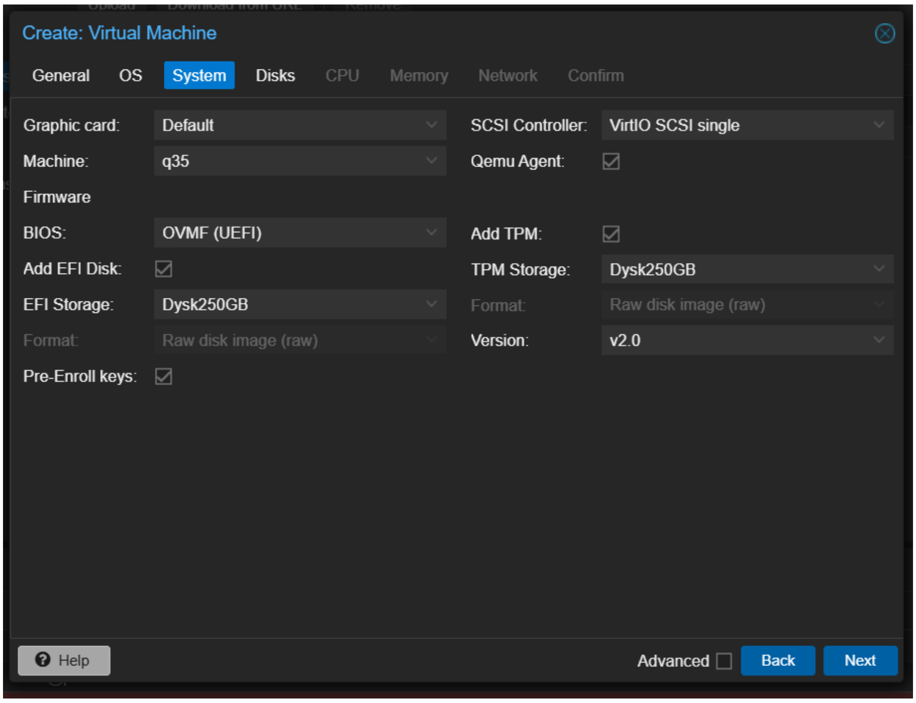


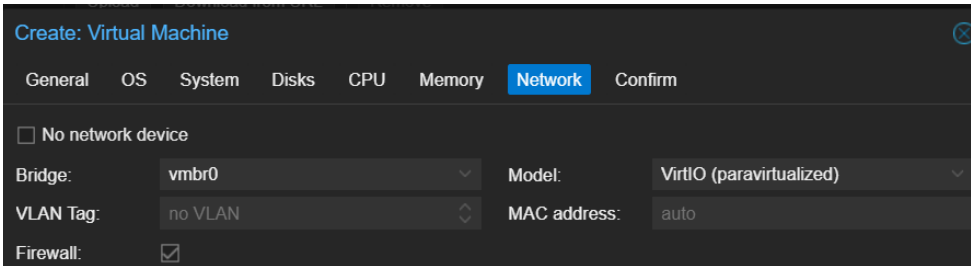


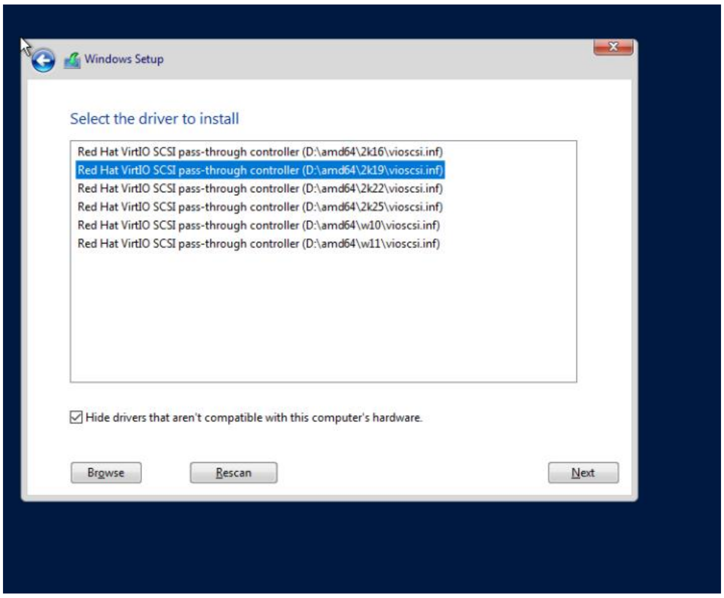
Zad 6.

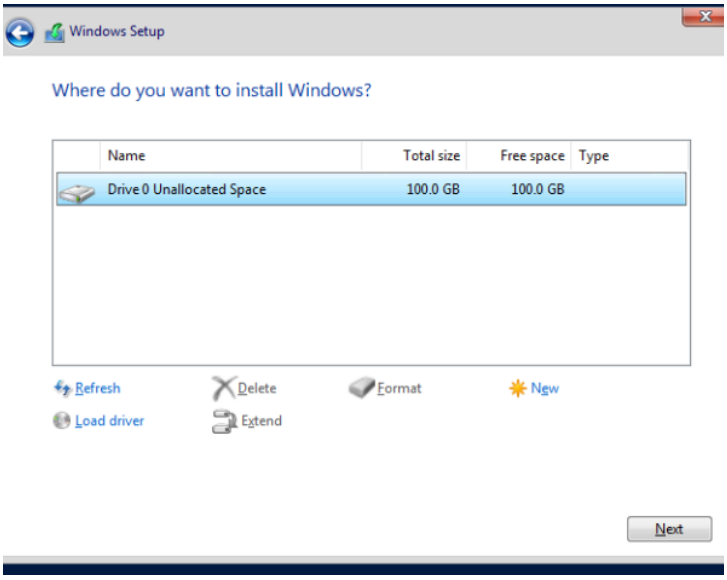
Optymalizacja Windows (VirtIO). Instalacja sterowników parawirtualnych, aby dyski i sieć w Windows działały z natywną prędkością hypervisora.

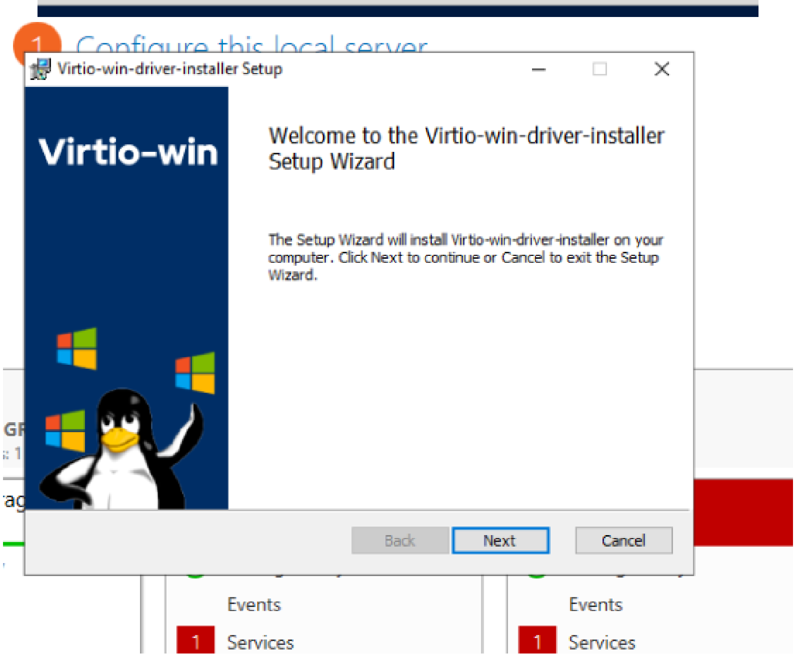






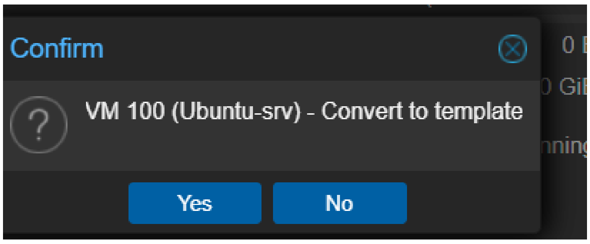

Windows Setup
Select the driver to install
Red Hat VirtIO SCSI pass-through controller (D:\amd64\2k16\vioscsi.inf)
Red Hat VirtIO SCSI pass-through controller (D:\amd64\2k19\vioscsi.inf)
Red Hat VirtIO SCSI pass-through controller (D:\amd64\2k22\vioscsi.inf)
Red Hat VirtIO SCSI pass-through controller (D:\amd64\2k25\vioscsi.inf)
Red Hat VirtIO SCSI pass-through controller (D:\amd64\w10\vioscsi.inf)
Red Hat VirtIO SCSI pass-through controller (D:\amd64\w11\vioscsi.inf)
Hide drivers that aren't compatible with this computer's hardware.
Browse
Rescan
Next

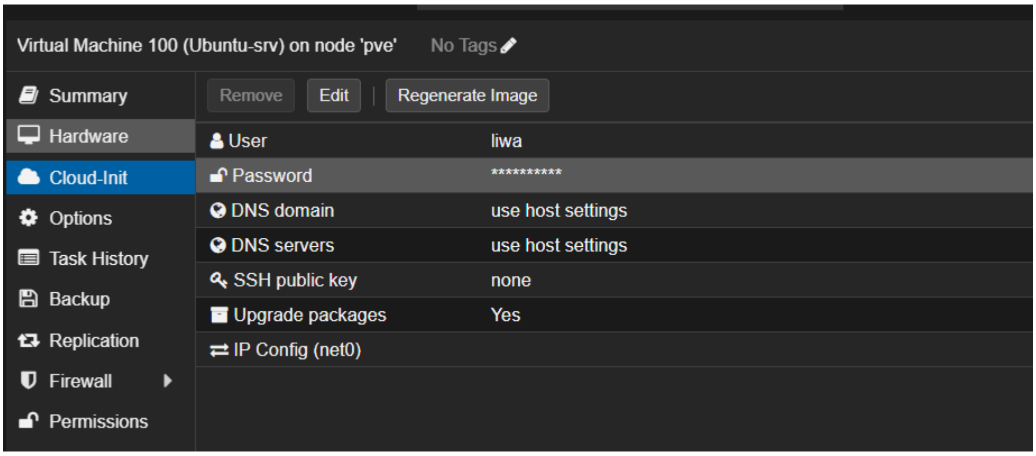

Windows Setup
Where do you want to install Windows?
Name
Total size
Free space
Type
Drive 0 Unallocated Space
100.0 GB
100.0 GB
Refresh
Delete
Format
New
Load driver
Extend
Next

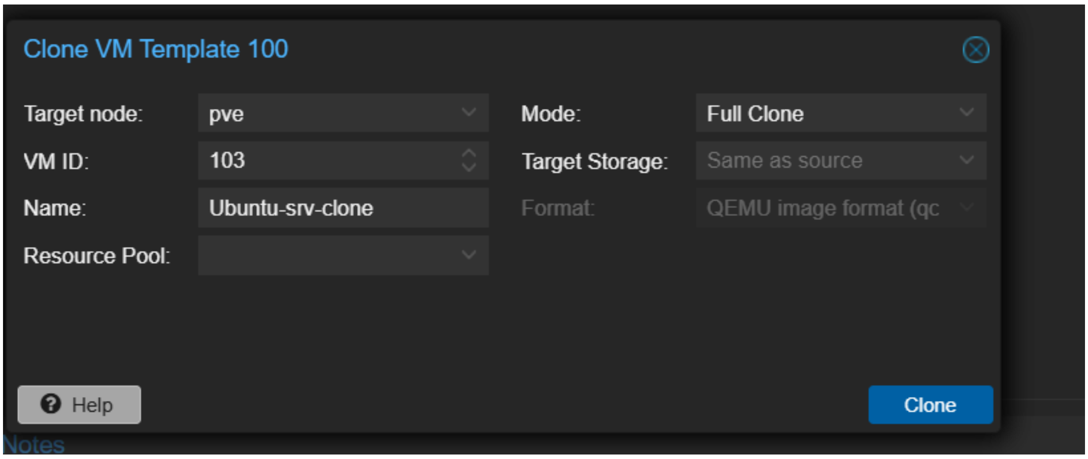

Configure this local server
Virtio-win-driver-installer Setup
Virtio-win
Welcome to the Virtio-win-driver-installer Setup Wizard
The Setup Wizard will install Virtio-win-driver-installer on your computer. Click Next to continue or Cancel to exit the Setup Wizard.
Back
Next
Cancel
Events
Services
Events
Services

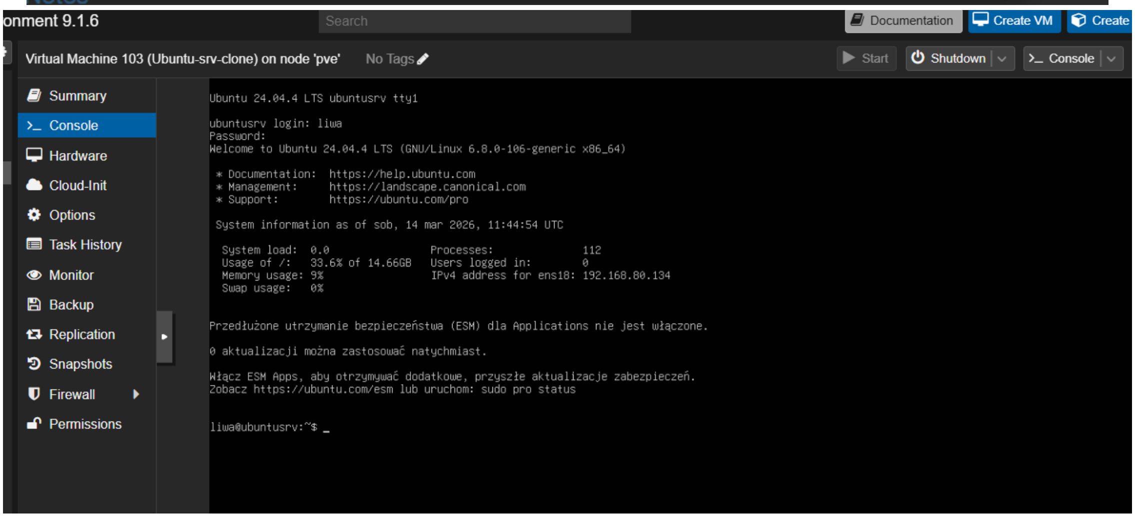
Zad 7.

Automatyzacja Linux (Template & Cloud-Init). Przygotowanie "czystego" obrazu Ubuntu (bez unikalnych ID), który sam skonfiguruje użytkownika przy każdym nowym wdrożeniu.



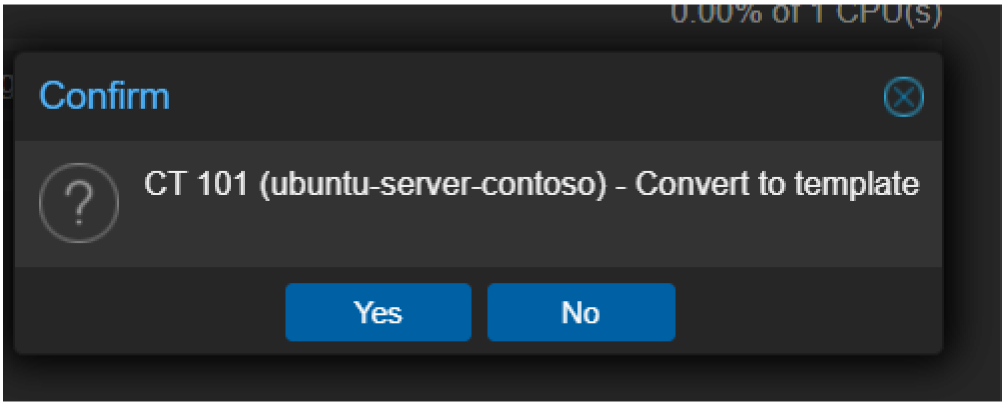


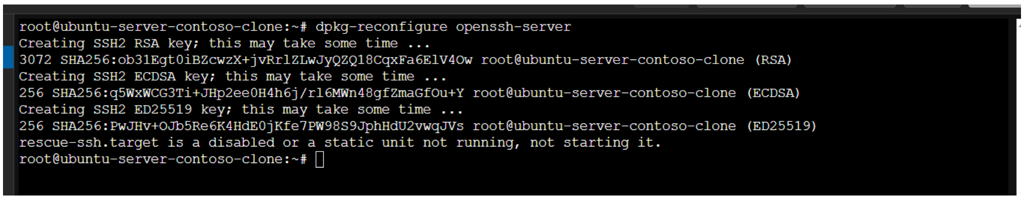




Zad 8.

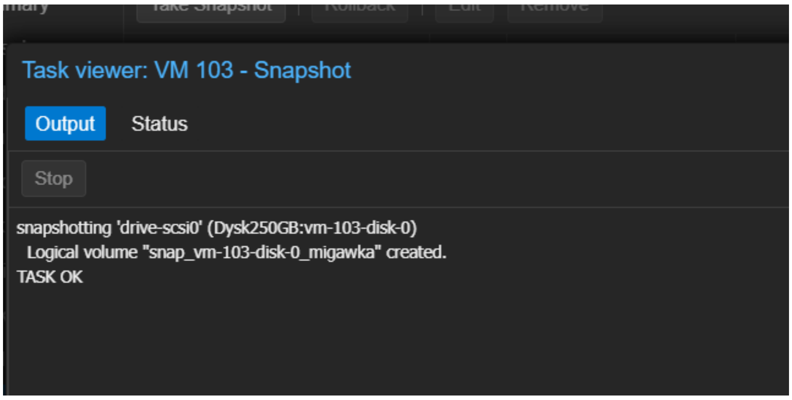
Szablony kontenerów (LXC). Szybkie klonowanie lekkich instancji systemowych i regeneracja kluczy bezpieczeństwa SSH po stronie klienta.

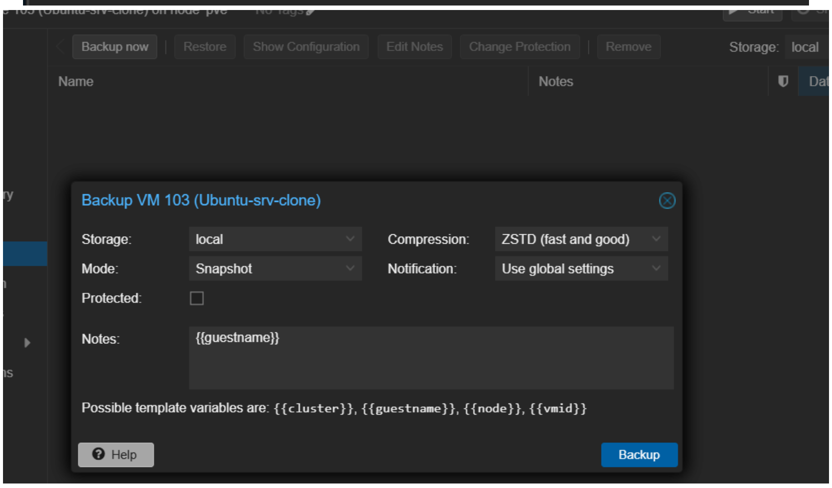


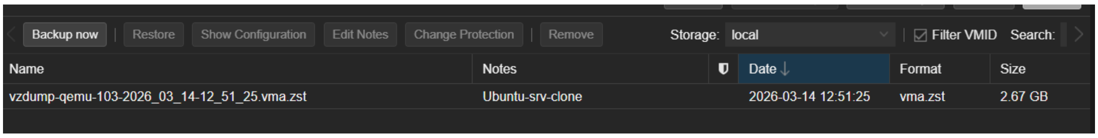


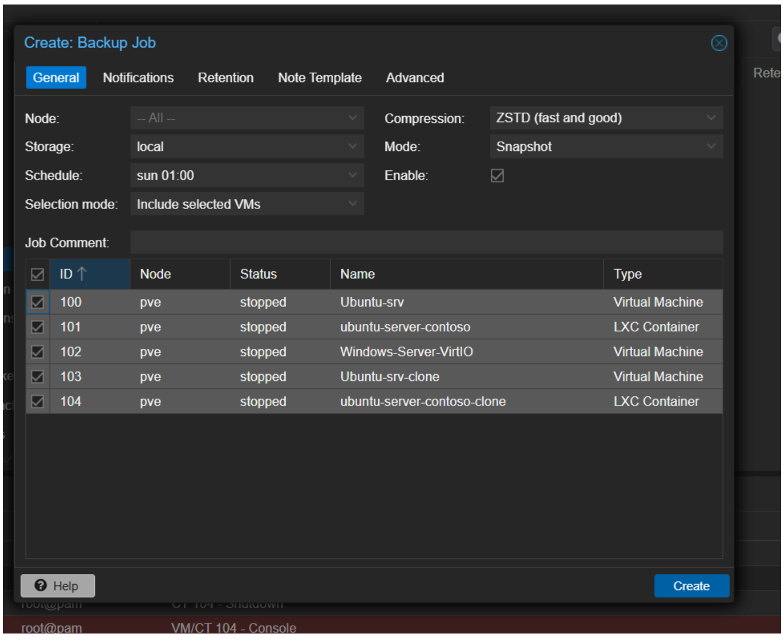
Zad 9.

Ciągłość biznesowa (Backup/Snapshot). Testowanie "punktów przywracania" przed zmianami oraz automatyzacja kopii zapasowych na wypadek awarii.







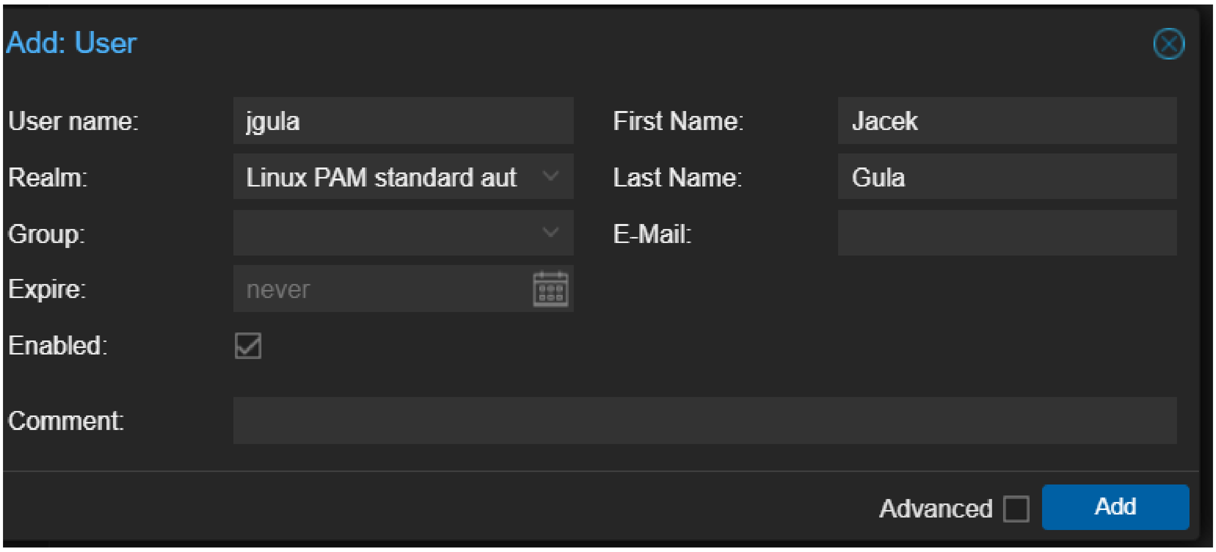


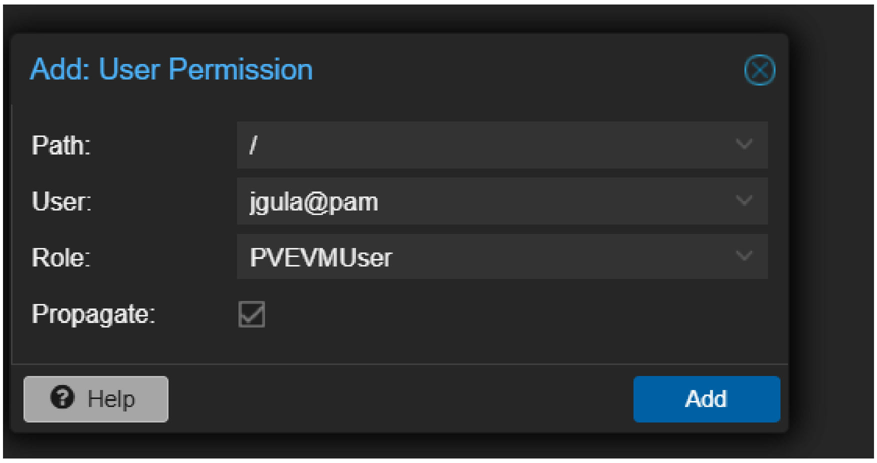
Zad 10.

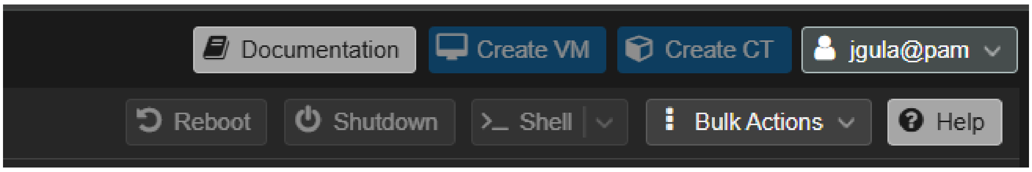
Izolacja dostępu (PAM). Tworzenie użytkownika systemowego, który zarządza Proxmoxem przez WWW, ale nie ma wstępu do konsoli serwera (bezpieczeństwo powłoki).

```
root@pve:~# adduser jgula
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for jgula
Enter the new value, or press ENTER for the default
        Full Name []: Jacek
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:
Is the information correct? [Y/n] Y
```

```
root@pve:~# usermod -s /bin/false jgula
```



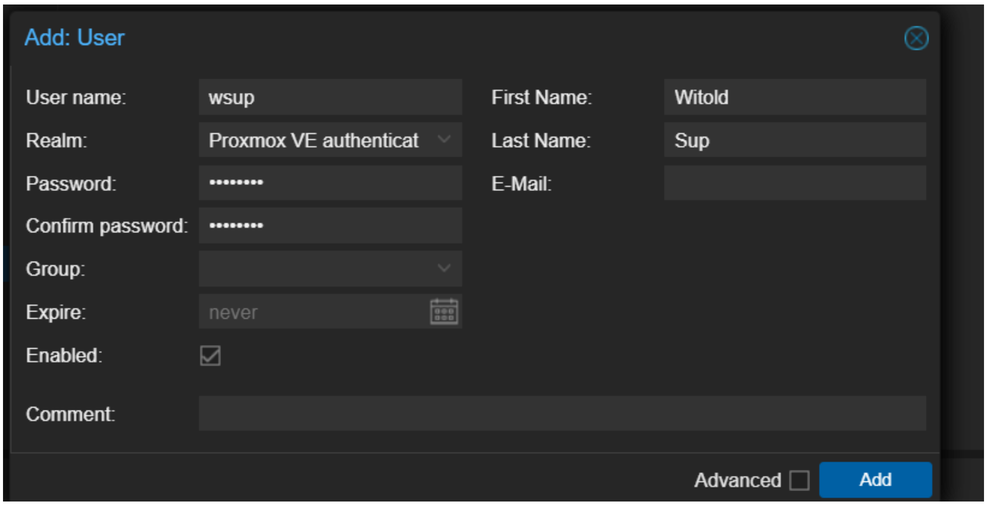


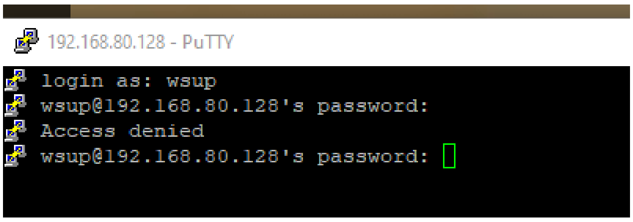


Brak dostępu do konsoli

Zad 11.

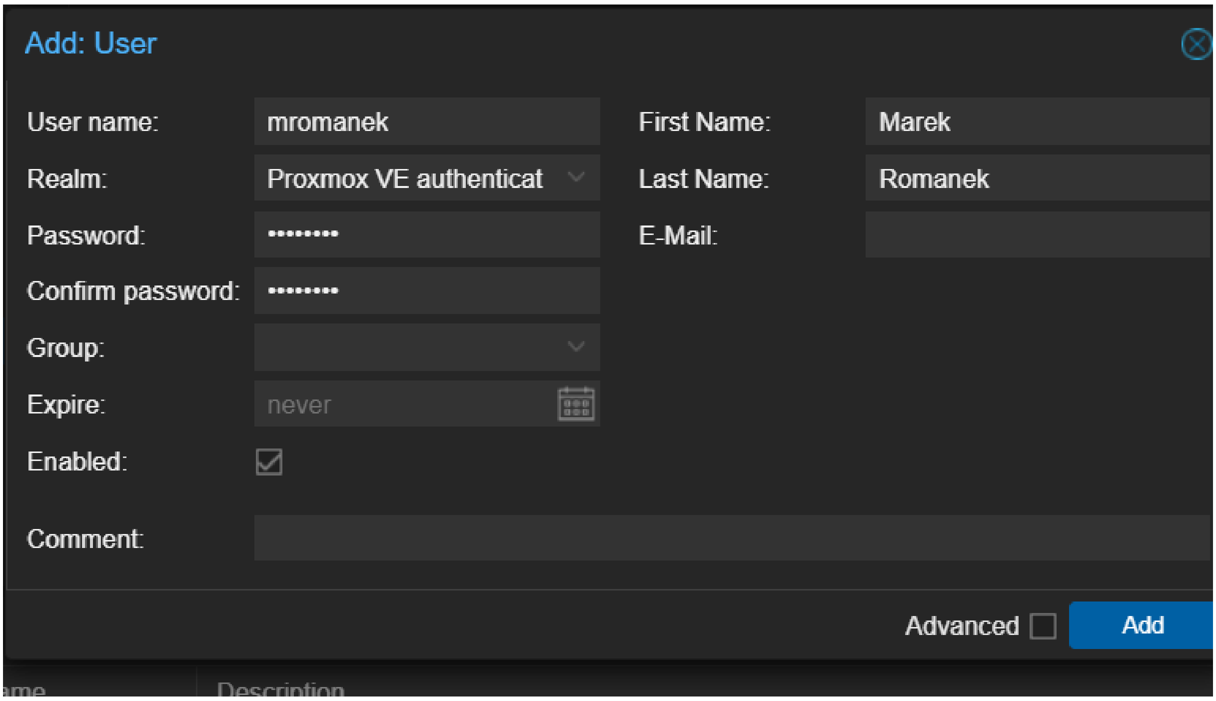
Uwierzytelnianie PVE. Testowanie różnicy między kontem systemowym (PAM) a kontem tylko wewnątrz bazy Proxmoxa (PVE).

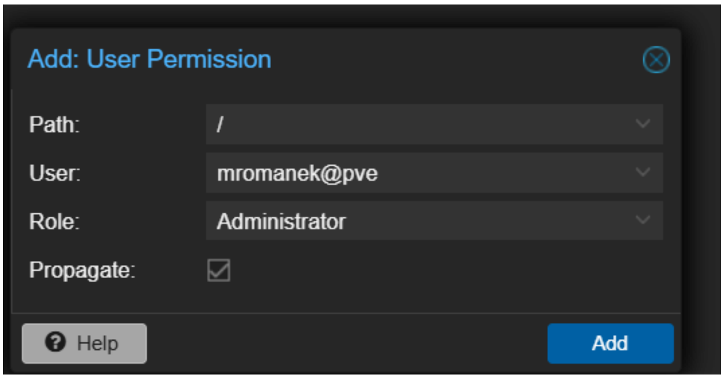


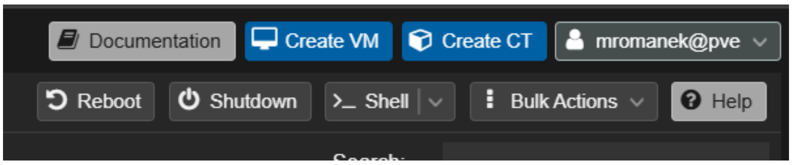


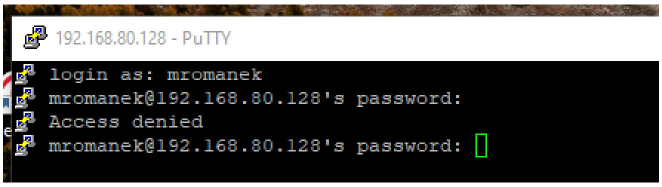
Zad 12.

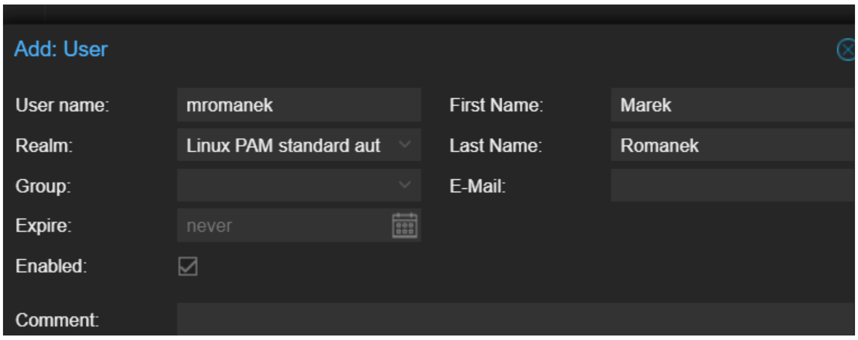
Testowanie uprawnień SSH. Weryfikacja, że tylko użytkownicy PAM mogą logować się bezpośrednio do systemu operacyjnego hosta.

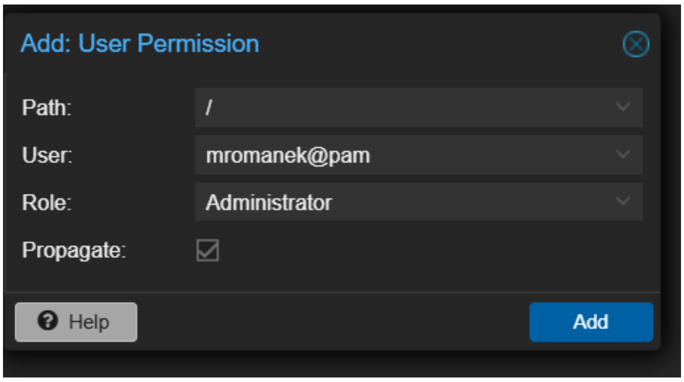


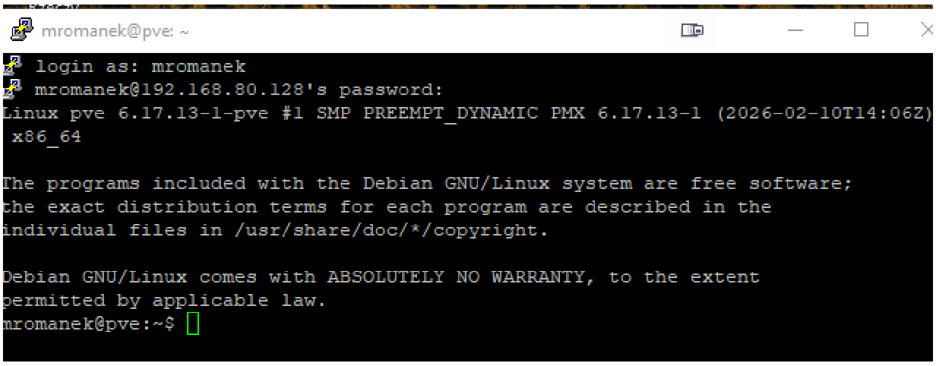

Add: User Permission
Path:
/
User:
mromanek@pve
Role:
Administrator
Propagate:
Help
Add


Documentation
Create VM
Create CT
mromanek@pve
Reboot
Shutdown
Shell
Bulk Actions
Help


192.168.80.128 - PuTTY
login as: mromanek
mromanek@192.168.80.128's password:
Access denied
mromanek@192.168.80.128's password:

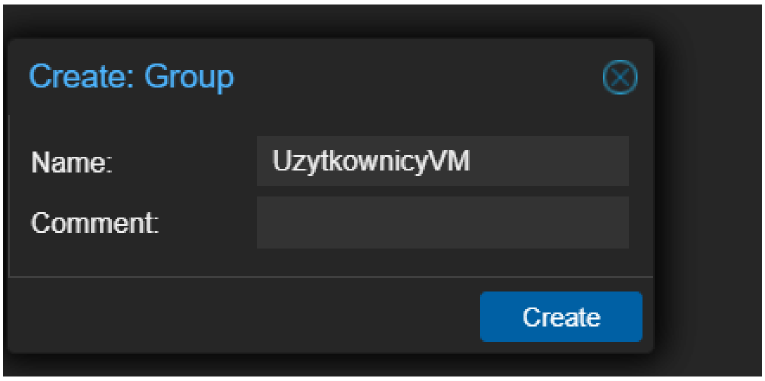

Add: User
User name:
mromanek
First Name:
Marek
Realm:
Linux PAM standard aut
Last Name:
Romanek
Group:
E-Mail:
Expire:
never
Enabled:
Comment:

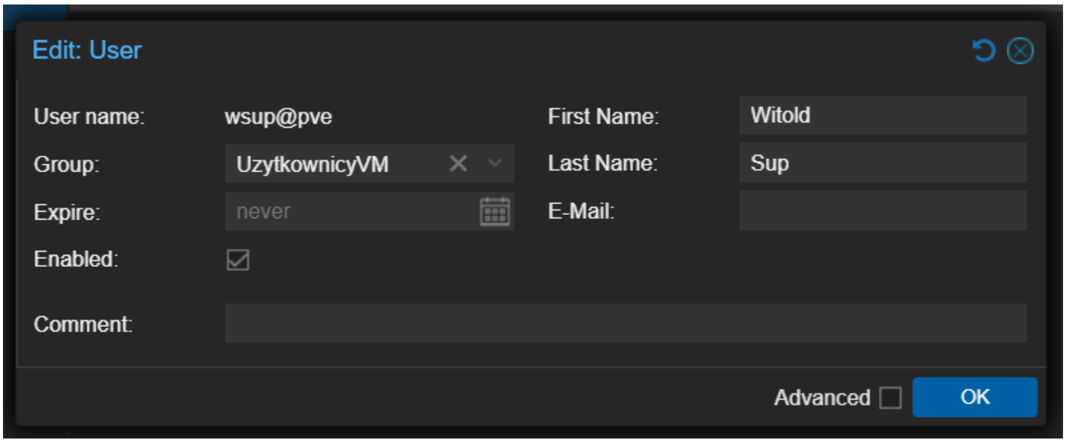

Add: User Permission
Path:
/
User:
mromanek@pam
Role:
Administrator
Propagate:
Help
Add

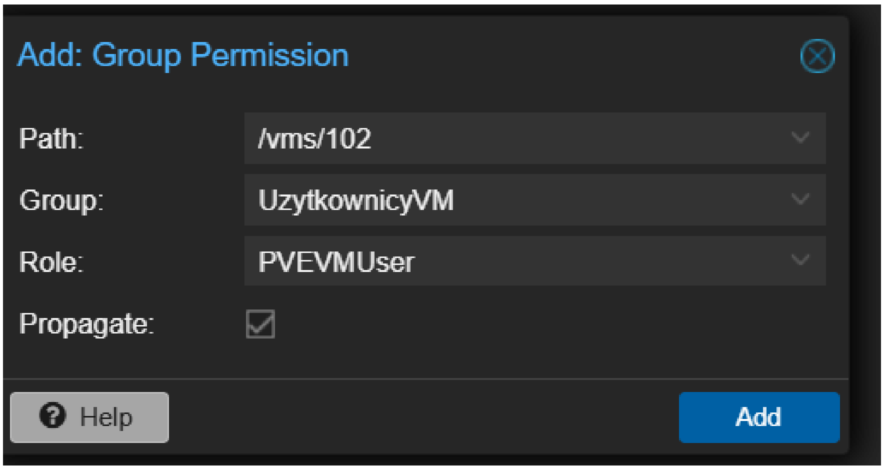

mromanek@pve: ~
login as: mromanek
mromanek@192.168.80.128's password:
Linux pve 6.17.13-1-pve #1 SMP PREEMPT_DYNAMIC PMX 6.17.13-1 (2026-02-10T14:06Z) x86_64
The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
mromanek@pve:~$

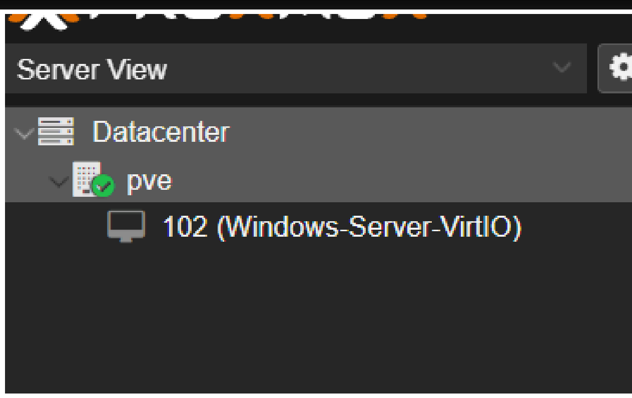
Zad 13.

Grupy i Rola PVEVMUser. Ograniczanie widoczności: użytkownik widzi i zarządza tylko konkretną maszyną (np. tylko Windows), a nie całym serwerem.



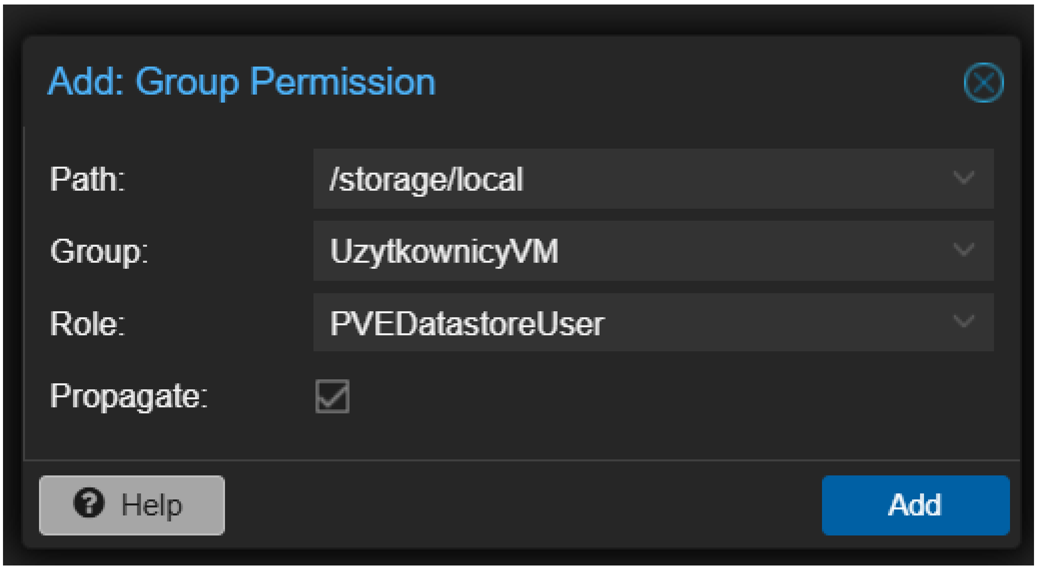


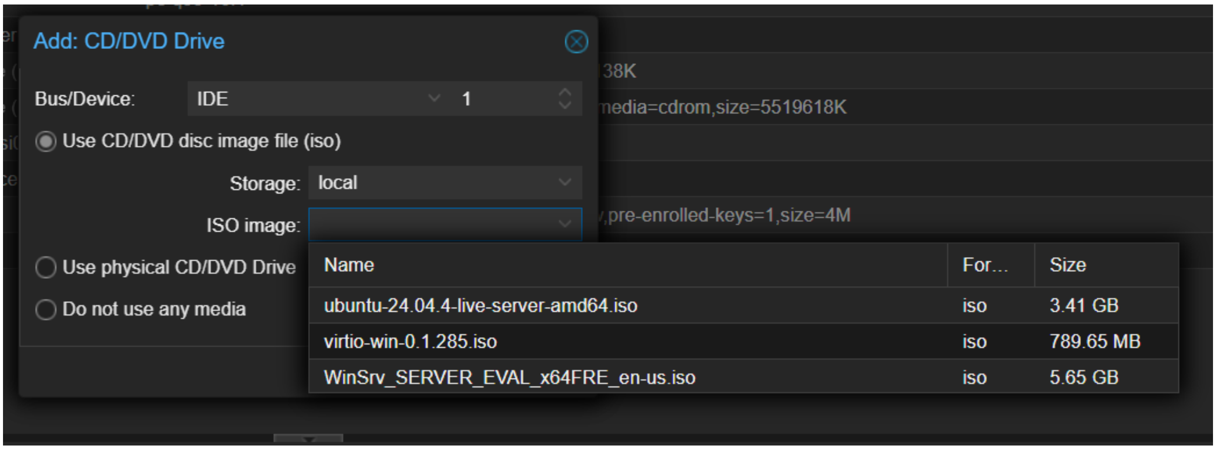




Zad 14.

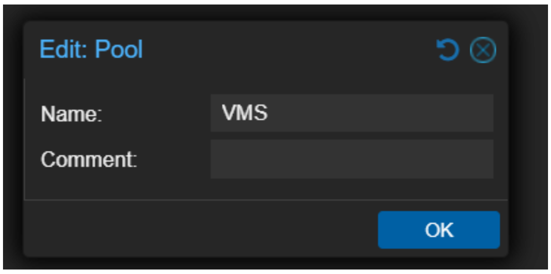
Dostęp do zasobów (Datastore). Nadawanie uprawnień do "magazynu", aby użytkownicy mogli samodzielnie montować obrazy ISO (płyty instalacyjne).

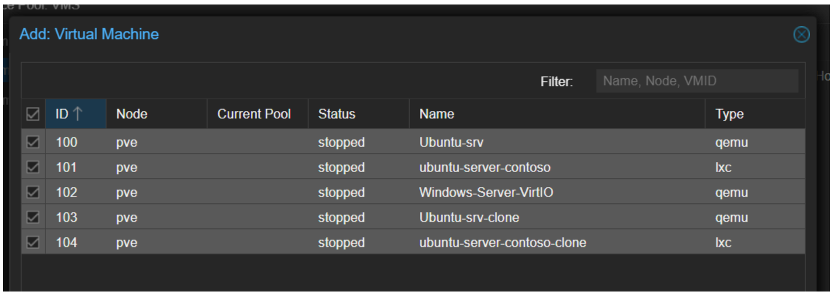


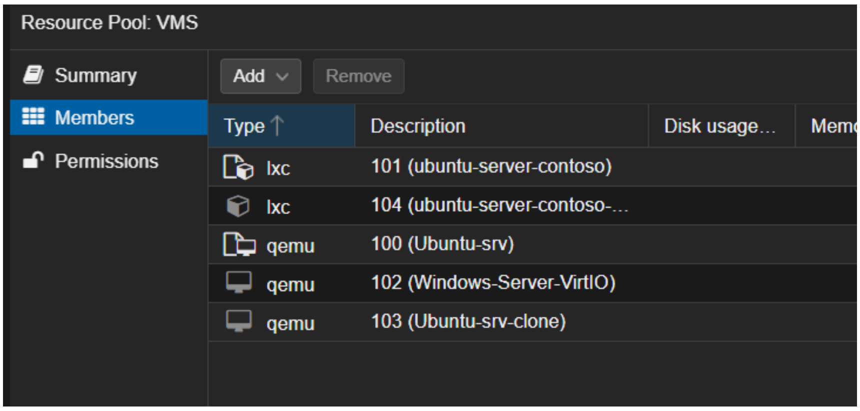


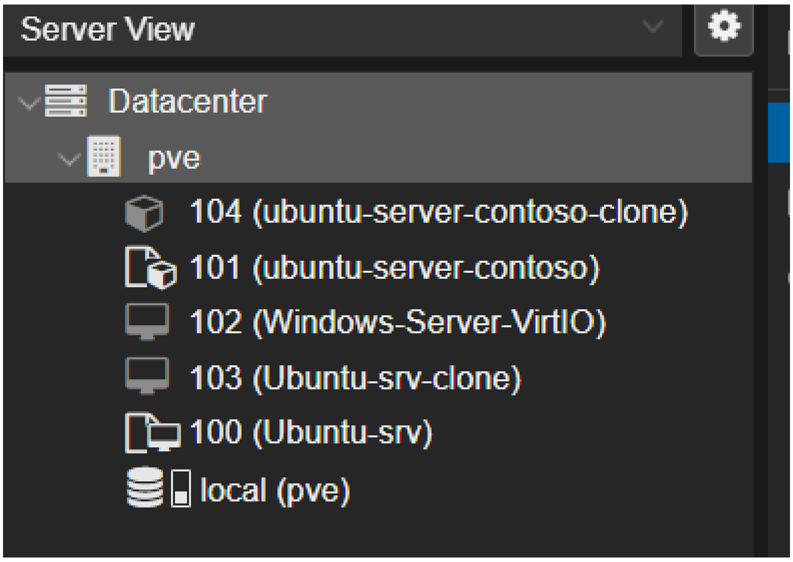
Zad 15.

Pule zasobów (Pools). Logiczne grupowanie maszyn w "worki", co pozwala nadać uprawnienia do wielu maszyn naraz jednym kliknięciem.



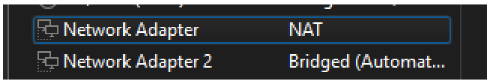






Zad 16.

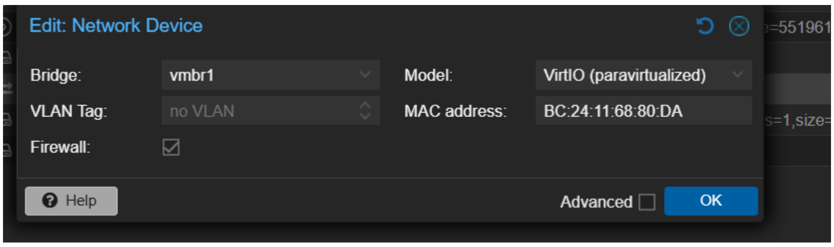
Passthrough/Dedicated NIC. Przypisanie fizycznej karty sieciowej wyłącznie dla jednej maszyny wirtualnej (bezpośredni dostęp do LAN).




```
root@pve:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: nic0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master vmbr0 state UP group default qlen 1000
    link/ether 00:0c:29:3f:14:a4 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c293f14a4
3: ens37: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 00:0c:29:3f:14:ae brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    altname enx000c293f14ae
4: vmbr0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:0c:29:3f:14:a4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.80.128/24 scope global vmbr0
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe3f:14a4/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
root@pve:~#
```

```
auto vmbr1
iface vmbr1 inet dhcp
        bridge-ports ens34
        bridge-stp off
        bridge-fd 0
```

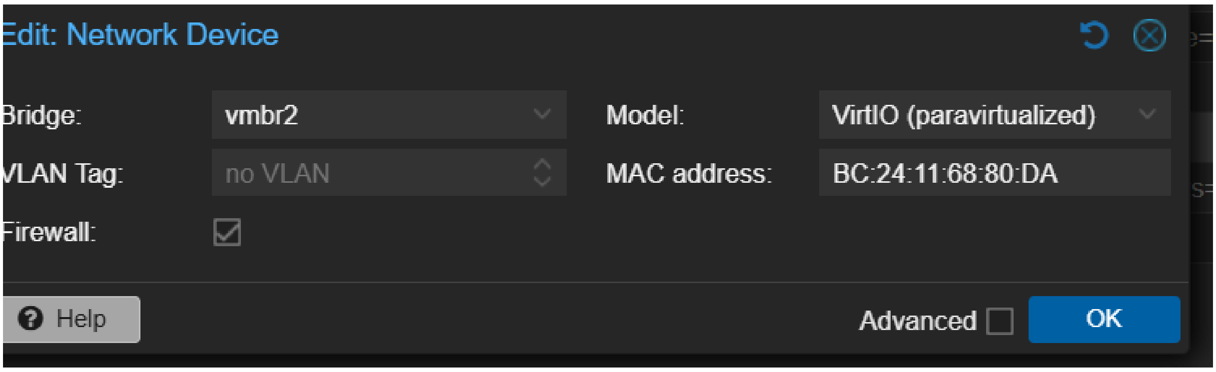
```
root@pve:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: nic0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel master vmbr0 state UP group default qlen 1000
    link/ether 00:0c:29:3f:14:a4 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c293f14a4
3: ens37: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 00:0c:29:3f:14:ae brd ff:ff:ff:ff:ff:ff
    altname enp2s5
    altname enx000c293f14ae
5: vmbr0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:0c:29:3f:14:a4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.80.128/24 scope global vmbr0
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe3f:14a4/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
6: vmbr1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UNKNOWN group default qlen 1000
    link/ether 86:1a:be:ff:d5:04 brd ff:ff:ff:ff:ff:ff
    inet6 fe80::841a:beff:feff:d504/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
root@pve:~#
```

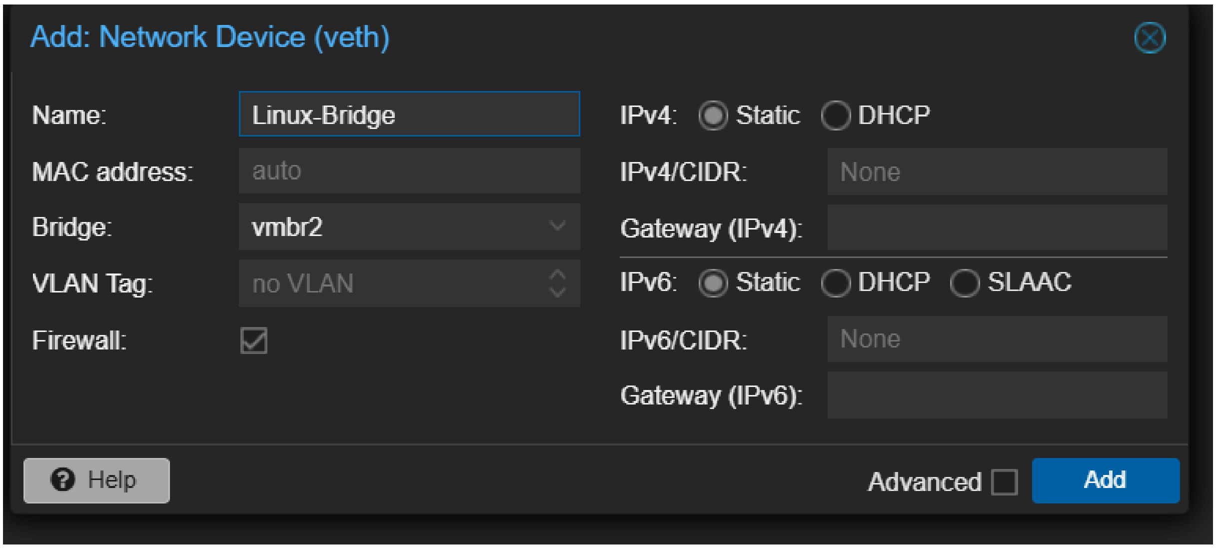


Zad 17.

Izolacja (Internal Network). Tworzenie wirtualnego switcha bez wyjścia na świat, aby maszyny rozmawiały tylko ze sobą (bezpieczny poligon).

```
auto vmbr2
iface vmbr2 inet static
        address 172.16.0.1/24
        bridge-ports none
        bridge-stp off
        bridge-fd 0
```





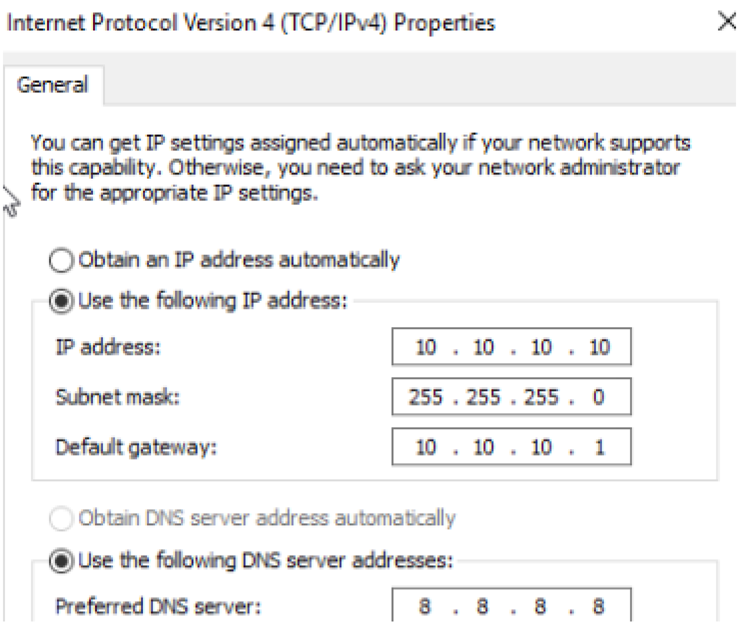

```
root@ubuntu-server-contoso-clone:~# ping 172.16.0.20
PING 172.16.0.20 (172.16.0.20) 56(84) bytes of data.
64 bytes from 172.16.0.20: icmp_seq=1 ttl=64 time=0.015 ms
64 bytes from 172.16.0.20: icmp_seq=2 ttl=64 time=0.036 ms
64 bytes from 172.16.0.20: icmp_seq=3 ttl=64 time=0.036 ms
64 bytes from 172.16.0.20: icmp_seq=4 ttl=64 time=0.036 ms
```

Zad 18.

Maskarada (NAT). Konfiguracja routera wewnątrz Proxmoxa, aby maszyny z prywatnej podsieci mogły pobierać dane z internetu.

```
auto vmbr3
iface vmbr3 inet static
        address 10.10.10.1/24
        bridge-ports none
        bridge-stp off
        bridge-fd 0

        post-up   echo 1 > /proc/sys/net/ipv4/ip_forward
        post-up   iptables -t nat -A POSTROUTING -s '10.10.10.0/24' -o eno1 -j MASQUERADE
        post-down iptables -t nat -D POSTROUTING -s '10.10.10.0/24' -o eno1 -j MASQUERADE
```

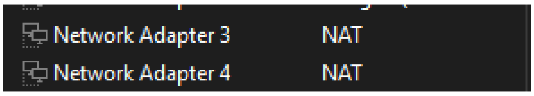
```
⇄ Network Device (net0)        virtio=BC:24:11:68:80:DA,bridge=vmbr3
```




```
C:\Users\Administrator>ping 10.10.10.1

Pinging 10.10.10.1 with 32 bytes of data:
Reply from 10.10.10.1: bytes=32 time<1ms TTL=64
Reply from 10.10.10.1: bytes=32 time<1ms TTL=64
Reply from 10.10.10.1: bytes=32 time<1ms TTL=64
Reply from 10.10.10.1: bytes=32 time<1ms TTL=64

Ping statistics for 10.10.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Zad 19.

Niezawodność (Bonding). Łączenie dwóch kart sieciowych w jedną (Active-Backup) – jeśli jedna się spali, sieć działa dalej bez przerwy.




```
auto bond0
iface bond0 inet dhcp
        slaves ens38 ens39
        bond-mode active-backup
        bond-miimon 100
        bond-primary ens38
```

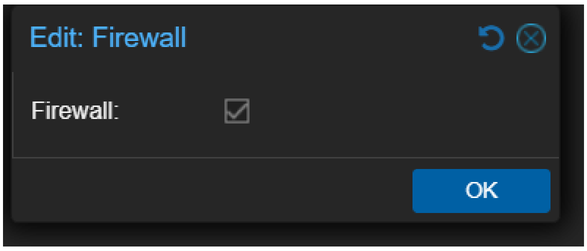
Zad 20.

Segmentacja (VLAN). Wpinanie maszyny wirtualnej w konkretny identyfikator VLAN (100), aby odseparować ruch sieciowy na poziomie warstwy 2.

```
auto vmbr2.100
iface vmbr2.100 inet dhcp
        vlan-raw-device ens40
```

Zad 21.

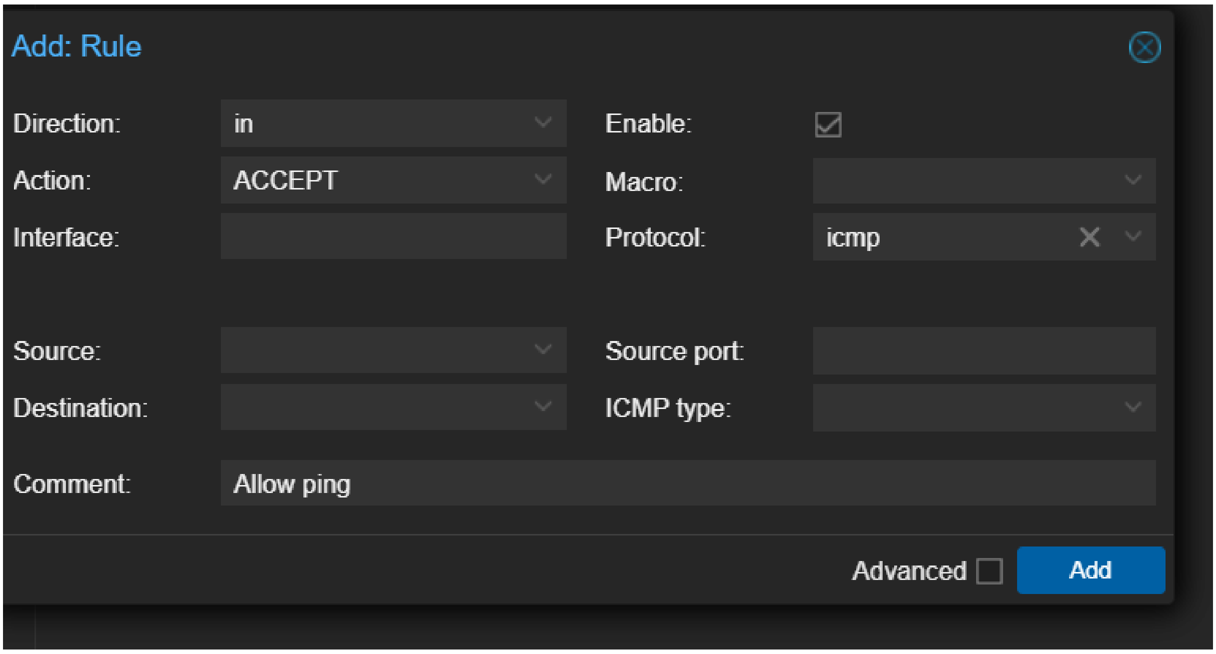
Ochrona Host’a. Aktywacja zapory na poziomie całego klastra i odblokowanie protokołu ICMP (Ping), aby monitorować dostępność serwera.




```
permitted by applicable law.
root@pve:~# cat /etc/pve/firewall/cluster.fw
[OPTIONS]

enable: 1
```

```
root@pve:~# iptables -Lvn
iptables: No chain/target/match by that name.
root@pve:~#
```

```
C:\Users\Admin>ping 192.168.80.128

Pinging 192.168.80.128 with 32 bytes of data:
Request timed out.
```

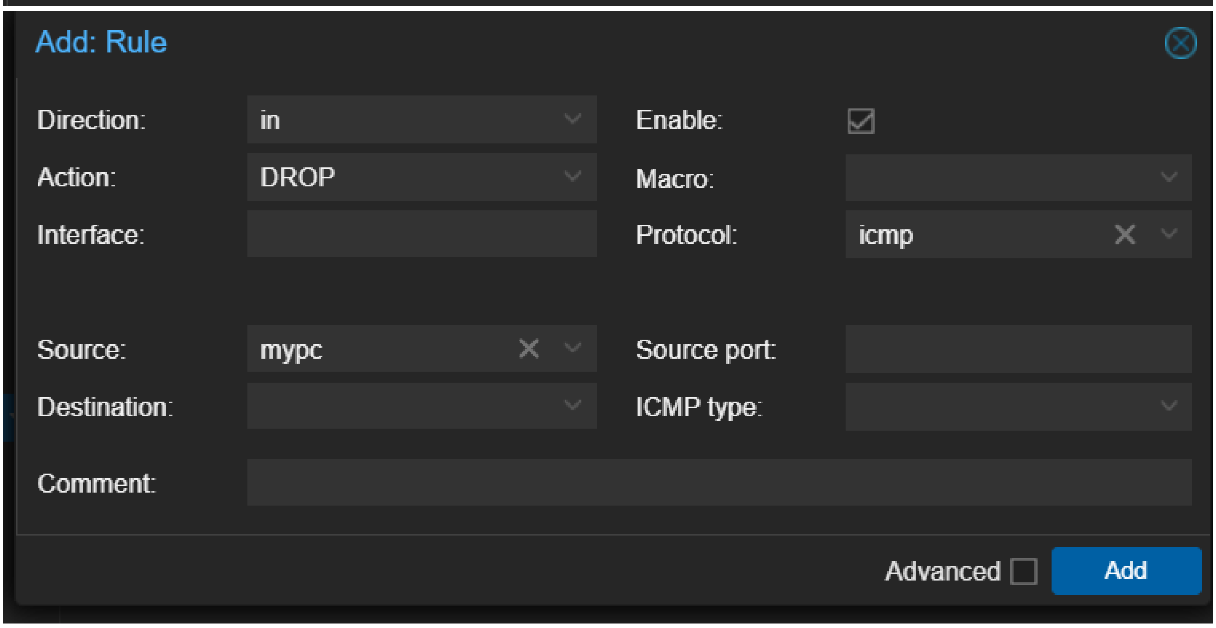


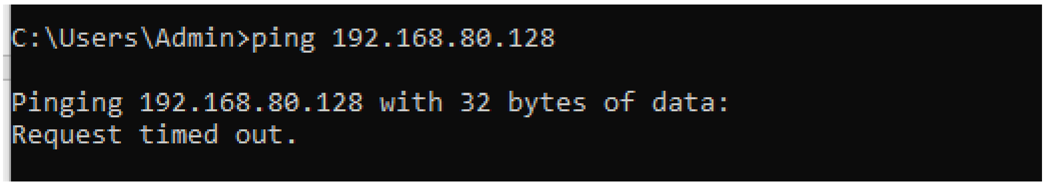

```
C:\Users\Admin>ping 192.168.80.128

Pinging 192.168.80.128 with 32 bytes of data:
Reply from 192.168.80.128: bytes=32 time<1ms TTL=64
Reply from 192.168.80.128: bytes=32 time<1ms TTL=64
```

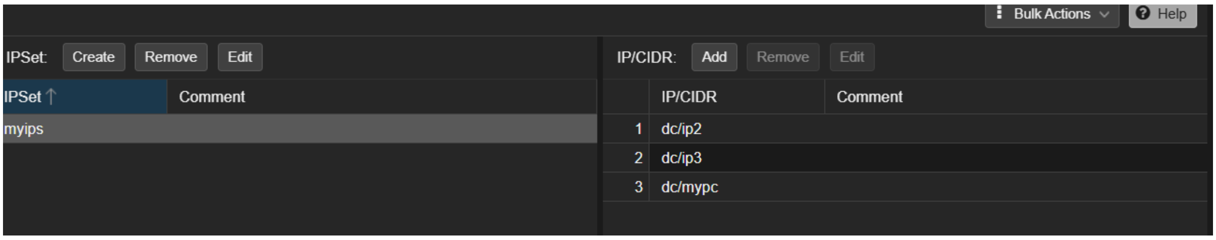
Zad 22.

Aliasy i IPSet. Grupowanie adresów IP w listy (IPSet), co pozwala jedną regułą blokować lub wpuszczać całe działy firmy.

Name	IP/CIDR ↑	Comment
myPC	192.168.80.1/24	myPC







Zad 23.

Ochrona Maszyn (VM Firewall). Precyzyjne otwieranie tylko niezbędnych portów (np. port 22 dla SSH) dla konkretnej wirtualnej maszyny.